

Module 1: reduced resistive-MHD state

The project physics is a strict two-dimensional, incompressible, doubly periodic reduced-MHD model:

$$\Omega = [0, 2\pi) \times [0, 2\pi), \quad t \in [0, T], \quad \partial_z = 0.$$

The primary scalar fields are magnetic flux and stream function,

$$\mathbf{q}(x, y, t) = (\psi, U),$$

with derived quantities

$$\mathbf{B}_\perp = \nabla\psi \times \hat{\mathbf{z}} = (\psi_y, -\psi_x), \quad \mathbf{v}_\perp = \hat{\mathbf{z}} \times \nabla U = (-U_y, U_x),$$

$$J = -\nabla^2\psi, \quad \omega = \nabla^2 U, \quad \nabla \cdot \mathbf{B}_\perp = \nabla \cdot \mathbf{v}_\perp = 0.$$

The surrogate receives a complete initial field pair and predicts the later field pair.

Module 1: brackets and governing equations

The two-dimensional bracket is

$$[f, g] = f_x g_y - f_y g_x, \quad \mathbf{v}_\perp \cdot \nabla f = [U, f].$$

Thus nonlinear advection appears compactly as a Poisson bracket. The frozen sign convention is

$$\boxed{\psi_t + [U, \psi] = \eta \nabla^2 \psi,}$$

$$\boxed{\omega_t + [U, \omega] = [J, \psi] + \nu \nabla^2 \omega,} \quad J = -\nabla^2 \psi, \quad \omega = \nabla^2 U.$$

Interpretation:

$[U, \psi]$: flux advection, $\eta \nabla^2 \psi$: resistive diffusion,
 $[J, \psi]$: magnetic forcing, $\nu \nabla^2 \omega$: viscous vorticity diffusion.

Module 1: scales, gauges, and energy

Alfven normalization gives

$$V_A = \frac{B_0}{\sqrt{\mu_0 \rho_0}}, \quad t_A = \frac{L_0}{V_A}, \quad \eta = \frac{\eta_m}{L_0 V_A}, \quad \nu = \frac{\nu_d}{L_0 V_A}.$$

Dimensionless control numbers are

$$S = \eta^{-1}, \quad \text{Re} = \nu^{-1}, \quad \text{Pm} = \frac{\nu}{\eta}.$$

Periodic gauges and means:

$$\langle U \rangle(t) = 0, \quad \frac{d}{dt} \langle \psi \rangle = 0, \quad \langle J \rangle = \langle \omega \rangle = 0.$$

The verification energy law is

$$\frac{d}{dt} \frac{1}{2} \int_{\Omega} (|\nabla \psi|^2 + |\nabla U|^2) dA = -\eta \int_{\Omega} J^2 dA - \nu \int_{\Omega} \omega^2 dA \leq 0.$$

Module 1: prediction task and claim boundary

The learned map is

$$(\psi_0, U_0, \eta, \nu, t) \mapsto (\psi_t, U_t).$$

The inputs have direct physical roles:

(ψ_0, U_0) : initial magnetic and flow state, (η, ν) : diffusion strengths,
 t : requested prediction time.

The outputs are complete numerical fields:

$$\hat{\psi}(x, y, t), \quad \hat{U}(x, y, t).$$

The physical model does not change when training changes from data-only to physics-informed.

Claim boundary

The project demonstrates a surrogate workflow for an independent two-field reduced-MHD model. It does not yet predict genuine M3D-C1 output.

Module 2: one simulation creates one data case

Choose the physical settings and initial fields:

$$(\psi_0, U_0, \eta, \nu).$$

The solver then uses the reduced-MHD equations from Module 1 to calculate

$$(\psi, U, J, \omega) \quad \text{at each saved time.}$$

One case file stores

$$x, y, t, \psi, U, J, \omega, \eta, \nu, A_0,$$

plus the initial conditions and solver settings.

Core idea

The saved numerical arrays are the training data. Plots are only pictures used to inspect those arrays.

Module 2: keep complete simulations together

Each saved time can become one learning example:

$$(\psi_0, U_0, \eta, \nu, t_k) \longrightarrow (\psi_k, U_k).$$

All times from one simulation stay together in training, validation, or testing. This prevents the same physical evolution from appearing on both sides of the comparison.

Before use, each case must pass simple-solution, grid, time-step, periodic-boundary, energy, and save/reload checks.

Claim boundary

These are reduced-MHD simulation data, not M3D-C1 results.

Module 3: one model contains the baseline and physics-informed versions

The neural network predicts

$$\mathcal{G}_\theta : (\psi_0, U_0, \eta, \nu, t) \mapsto (\hat{\psi}, \hat{U}).$$

The data loss compares these predictions with Module 2:

$$\mathcal{L}_{\text{data}} = \langle (\hat{\psi} - \psi)^2 \rangle + \langle (\hat{U} - U)^2 \rangle.$$

The baseline is simply

$$\mathcal{L}_{\text{total}} = w_d \mathcal{L}_{\text{data}}.$$

Why the separate baseline module disappeared

The physics-informed version uses the same inputs, outputs, and model. It only turns on additional loss terms.

Module 3: magnetic-flux residual

Move the predicted magnetic-flux equation to the left:

$$R_\psi = \hat{\psi}_t + [\hat{U}, \hat{\psi}] - \eta \nabla^2 \hat{\psi}.$$

The target is $R_\psi = 0$ at every location and time:

$$\mathcal{L}_{\text{PDE},\psi} = \left\langle \left(\frac{R_\psi}{S_\psi} \right)^2 \right\rangle.$$

Physical balance:

$$\underbrace{\hat{\psi}_t}_{\text{time change}} + \underbrace{[\hat{U}, \hat{\psi}]}_{\text{flow transports flux}} - \underbrace{\eta \nabla^2 \hat{\psi}}_{\text{resistive diffusion}} = 0.$$

This term asks for more than a similar-looking field: its time change, advection, and diffusion must balance.

Module 3: vorticity residual

First derive current and vorticity from the predicted fields:

$$\hat{J} = -\nabla^2 \hat{\psi}, \quad \hat{\omega} = \nabla^2 \hat{U}.$$

Then form

$$R_\omega = \hat{\omega}_t + [\hat{U}, \hat{\omega}] - [\hat{J}, \hat{\psi}] - \nu \nabla^2 \hat{\omega}.$$

and penalize

$$\mathcal{L}_{\text{PDE}, \omega} = \left\langle \left(\frac{R_\omega}{S_\omega} \right)^2 \right\rangle.$$

This enforces balance among vorticity change, flow advection, magnetic forcing, and viscous diffusion. The two PDE losses remain separate so that one equation cannot hide the other.

Module 3: initial, boundary, and gauge constraints

The initial state selects the correct solution of the PDE:

$$\mathcal{L}_{\text{IC}} = \left\langle (\hat{\psi}(0) - \psi_0)^2 \right\rangle + \left\langle (\hat{U}(0) - U_0)^2 \right\rangle.$$

The stream-function gauge removes an arbitrary constant:

$$\mathcal{L}_{\text{gauge}} = \left\langle \hat{U} \right\rangle^2.$$

For a periodic Fourier representation,

$$\boxed{\mathcal{L}_{\text{BC}} = 0}$$

because periodicity is built in. Two other constraints are also automatic:

$$\nabla \cdot \hat{\mathbf{B}}_{\perp} = 0, \quad \nabla \cdot \hat{\mathbf{v}}_{\perp} = 0,$$

because the fields are constructed from $\hat{\psi}$ and $\hat{U}$.

Module 3: final proof-of-concept objective

The recommended first physics-informed loss is

$$\mathcal{L}_{\text{total}} = w_d \mathcal{L}_{\text{data}} + w_\psi \mathcal{L}_{\text{PDE},\psi} + w_\omega \mathcal{L}_{\text{PDE},\omega} + w_0 \mathcal{L}_{\text{IC}} + w_g \mathcal{L}_{\text{gauge}}.$$

Optional tests can add

$$\mathcal{L}_{\text{mean}}, \quad \mathcal{L}_{\text{energy}}, \quad \mathcal{L}_{J,\omega}.$$

The weights prevent one numerically large term from overwhelming the others.

What this presentation establishes

It shows exactly how simulation agreement, the two MHD equations, and known conditions are combined to construct a physics-informed surrogate. Accuracy or speed improvements require later trained results.